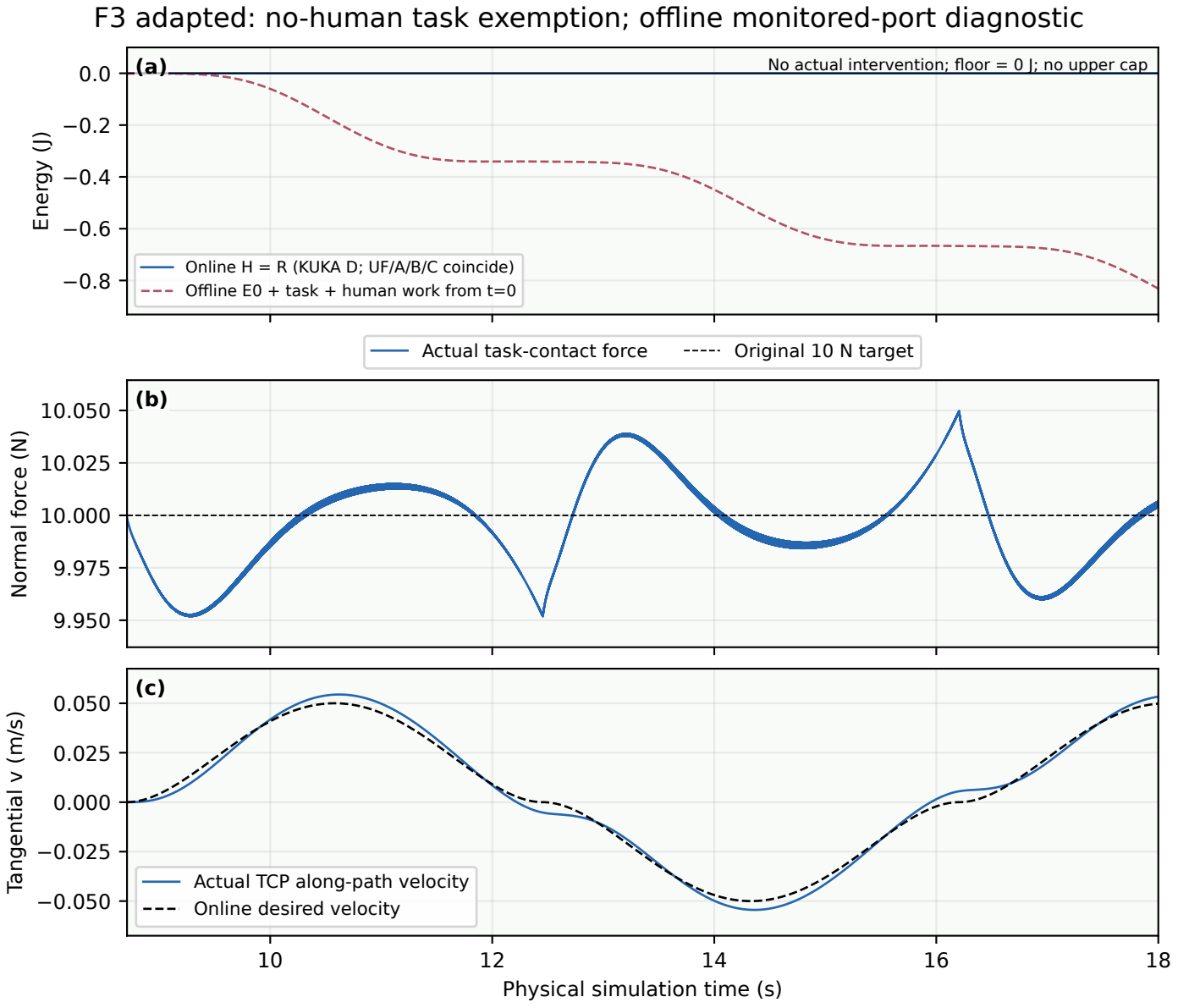

F3 adapted: no-human task exemption; offline monitored-port diagnostic
(a)
No actual intervention; floor = 0 J; no upper cap
Energy (J)
0.0
−0.2
−0.4
−0.6
−0.8
Online H = R (KUKA D; UF/A/B/C coincide)
Offline E0 + task + human work from t=0
Actual task-contact force
Original 10 N target
(b)
Normal force (N)
10.050
10.025
10.000
9.975
9.950
(c)
Tangential v (m/s)
0.050
0.025
0.000
−0.025
−0.050
Actual TCP along-path velocity
Online desired velocity
10
12
14
16
18
Physical simulation time (s)